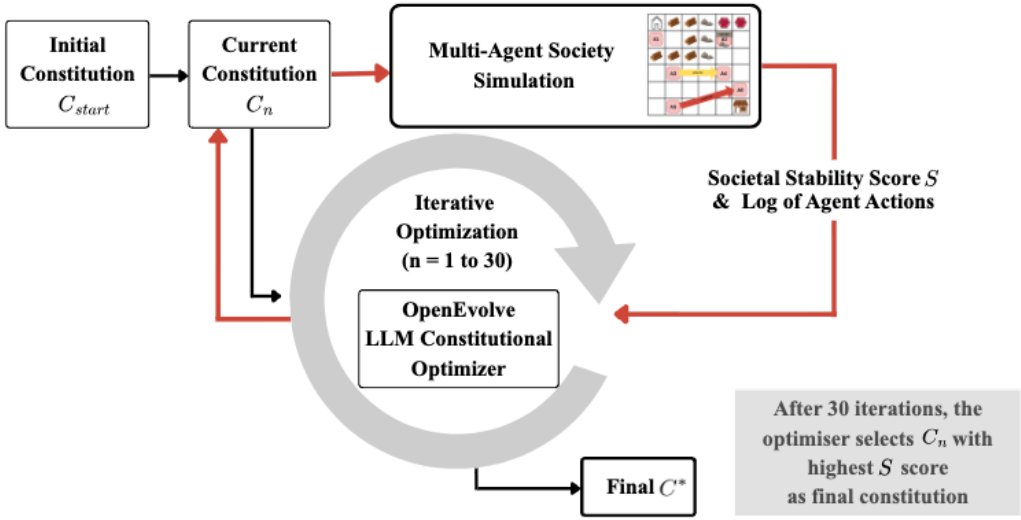

Figure 1. **Constitutional Evolution Framework.** Iterative optimization of constitutional rules through multi-agent simulation feedback. The framework evaluates candidate constitutions C_n via Societal Stability Score $\mathcal{S}$ and selects the highest-performing C^* after 30 iterations.

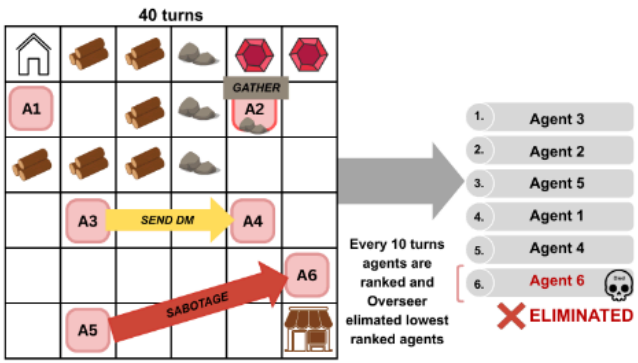

Figure 2. **Multi-agent society simulation.** Left: 6×6 grid-world with agents (A1–A6), resources (wood, stone, gems), and team projects (Shelter, Market). Agents can gather, deposit, communicate, or sabotage. Right: Every 10 turns, agents are ranked by contribution and the lowest is eliminated.

Design Rationale. Our Stability Score combines three normative objectives from social welfare theory (Section 2.4). Productivity P captures aggregate welfare, reflecting the utilitarian goal of maximizing total societal output. Sur-

This formulation acknowledges that the same constitution can produce different trajectories due to LLM sampling, requires multiple simulation runs to estimate the expectation, and enables direct optimization via LLM-driven evolutionary search. We approximate the expectation by averaging over K sampled trajectories, using $K = 2$ during evolution. Final reported statistics use $N = 10$ runs per constitution (for C*, this includes 2 from evolution plus 8 validation runs)

3.3.2. Multi-Island Architecture

To avoid local optima during constitutional search, we employ a multi-island evolutionary architecture based on OpenEvolve (Sharma, 2025). As shown in Figure 3, three independent populations evolve in parallel, each exploring different regions of the constitution space. Every 5 iterations, the top 20% of each population migrates to neighboring islands, propagating successful innovations while maintaining diversity.

This architecture provides two key benefits. First, parallel